CHICAGO BOOTH

The University of Chicago Booth School of Business

Business 41204 – Machine Learning

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Overview

This course aims to offer a broad, relatively non-technical introduction to key ideas from machine learning without digging too far into the mathematical minutiae. I intend the course to be accessible to students with an understanding of basic statistical concepts and mathematics but no deeper training beyond standard MBA-level statistics. Overall, my goal is that students leave the course able to have informed conversations about machine learning, understanding key conceptual ideas and practical challenges, and having a better idea of realistic opportunities for the use of machine learning in business.

The course will start by covering two basic learning paradigms: (i) supervised learning and (ii) unsupervised learning. In class, I will illustrate the paradigms through small-scale applications styled to match business problems. These applications will provide a vehicle for discussing important concepts and practical considerations. After covering the basic learning methods, we will explore additional topics such as deep neural networks, causal inference, policy learning, reinforcement learning, and generative AI models.

Prerequisites

BUS 41000 (or equivalent). Prerequisite material includes fundamentals of probability and statistics as taught in BUS 41000. I will assume that students are familiar with descriptive statistics (e.g. sample mean, standard deviation, covariance/correlation), random variables, basic probability calculations, normal distributions, confidence and prediction intervals, sampling distributions, hypothesis testing, and the basics of linear regression.

Students with previous background in statistics or data science from work or courses outside of Booth are welcome to take the class. It is the student's responsibility to self-assess having sufficient background and to withdraw early in the quarter if it becomes clear that the course is not a good fit.

Importantly, this course does not cover basic statistics material that is assumed for other courses at Booth. Such classes may not accept Machine Learning as a replacement for 41000 or other prerequisites.

Software

The use of software is an important element of the machine learning toolkit. I will use Python for in-class examples. Successful completion of course requirements will also require that students complete problem sets and a course project. Writing and executing code will be part of these course requirements. A simple recommendation is for students to run Python from notebooks within the Google Colab environment or to make use of other AI-based solutions that allow code writing and execution. Students following this recommendation will be able to complete course requirements without downloading, installing, or maintaining any software. To this end, problem sets will include starter notebooks suitable for running in Google Colab or similar environments. I expect that students without coding background will use vibe coding as part of their approach to completing course requirements. That is, students are not expected to already know how to code, but they are expected to actively engage with code, whether directly or with the assistance of AI tools.

AI Tools

I see AI tools (e.g. ChatGPT, Claude, Gemini, …) as productivity enhancing, and believe a machine learning course is a great environment to experiment with such tools. I strongly encourage students to make use of AI tools as they complete problem sets, work on the course project, and review material more generally.

Text and Class Notes

The lecture notes (available on Canvas) cover everything you need to know – in outline form – for this course.

There is no required textbook. The book [Introduction to Statistical Learning with Applications in Python] by James, Witten, Hastie, Tibshirani, and Taylor (ISL) is a relatively approachable text that covers supervised and unsupervised learning methods. It is available as a free .pdf download. I recommend ISL for students who want to dig a bit deeper into a textbook treatment of the supervised and unsupervised learning topics we cover. Machine Learning for Business Analytics: Concepts, Techniques, and Applications in Python (2nd Edition) by Shmueli, Bruce, Gedeck, and Patel covers approximately the same material as we go over in this course, plus additional topics.

I may assign readings throughout the course at my discretion.

Homework and Exams

I will assign course grades based on in-class participation (30%), group problem sets (including one presentation) (40%: 30% for problem set write-ups, 10% for presentation), and a group project and presentation (30%). Grades will be assigned based on a curve.

Note that Booth policy mandates that the GPA average within the course cannot exceed 3.33 (B+).

Details:

Group Work: I believe collaborative work is an important element of business education. As such, *a course requirement* is that students complete problem sets and a course project within groups. The plan is that groups will consist of 6 students, with minor adjustments to deal with integer problems. Students may choose their own groups, but any student that is not part of a study group by the end of the first course week (March 29, 11:59 PM CT) will be assigned to a group with less than five members.

- *Problem Sets* – Problem sets will be assigned throughout the quarter. The problems are meant to introduce practical application of machine learning methods, review concepts from the course, and provide a mechanism to think about important concepts from the course. Each group should submit only one write-up.

- *Project* – In addition to problem sets, each group will complete a course project. The course project will involve researching a topic related to machine learning that we do not directly cover in the course material. The project is relatively open-ended. We will provide example topics, but students are welcome to choose other topics and should work on something that their homework group finds interesting. The project does not need to involve substantive data analysis or technical work, though that is certainly an option. Details about the course project are available in a separate project document on Canvas.

- *Presentations* – As with the ability to work collaboratively, I believe a key element to success is the ability to distill and evaluate information, including information obtained from generative AI, and then convey the key takeaways to other humans. As such, I will require each group to give two presentations during the course. In course weeks 4-8 (April 13-May 11), groups will give presentations based on the problem sets. Each group must present once during this period. Each group will also give a presentation based on their course project in the final course week (week 9, May 18). The professor, the TAs, and all members of the course will evaluate every presentation by completing a brief survey following the presentation. Failure to complete an evaluation of any presentation will result in a penalty to your participation grade.

- *Group Peer Evaluation* – Each group member must complete an online group peer evaluation survey after the quarter is completed. The link to this survey will become active on Canvas at the end of the course. This evaluation prevents problems of free-riding by group members. Each member is required to submit an "effort" rating (0-100%) for each member of their group. The average rating across all members will be taken as the final "effort" rating for a group member. A 90% rating implies that the group member will get 90% of the group grade (the combination of group problem set and project grades). Failure to submit peer evaluation scores will result in a penalty to your participation grade.

Course Participation – Students actively engaging during lectures and with their peers enhances the learning environment. To encourage meaningful engagement, there is a course participation component to grades. Course participation is scored on a 120-point scale and then mapped into the 30% participation component of the final course grade.

All students start with a 90/120 in course participation. Attending each lecture, completing course requirements (including peer evaluations), but otherwise not engaging will result in students maintaining a 90/120 for course participation. Failure to attend a lecture will result in a reduction of 10 course participation points. Failure to complete an evaluation of a peer presentation will result in a reduction of 5 course participation points. Failure to complete the end-of-course evaluation of your study group peers will result in a loss of 45 course participation points. Negative course participation scores will result in lowered grades beyond just getting 0 for course participation. Extremely insightful questions, comments, or other engagement in the course will result in adding course participation points at the TAs’ and instructor’s discretion. The anticipation is that the majority of students will end the quarter with 90/120 for course participation.

Accommodations for Disabilities

The University of Chicago is committed to ensuring the full participation of all students in its programs. If you have a documented disability (or think you may have a disability) and, as a result, need a reasonable accommodation to participate in class, complete course requirements, or benefit from the University's programs or services, please contact Student Disability Services as soon as possible. To receive a reasonable accommodation, you must be appropriately registered with Student Disability Services. Please contact the office at 773-702- 6000/TTY 773-795-1186 or disabilities@uchicago.edu, or visit the website at disabilities.uchicago.edu. Student Disability Services is located at 5501 S. Ellis Avenue.

Tentative Schedule

Note that this schedule is tentative and subject to change.

Week 1 (23 Mar):	Supervised Learning 1
Week 2 (30 Mar):	Supervised Learning 2
Week 3 (6 Apr):	Unsupervised Learning
Week 4 (13 Apr):	Deep Neural Networks Group presentations
Week 5 (20 Apr):	Causal Inference Group presentations
Week 6 (27 Apr):	Policy Learning Group presentations
Week 7 (4 May):	Introduction to Foundation Models Group presentations
Week 8 (11 May):	SLACK/TBD Group presentations
Week 9 (18 May):	Final project presentations